Interactive Ice-Shape Probing and Control for Grid-based CFD

SmaggIce

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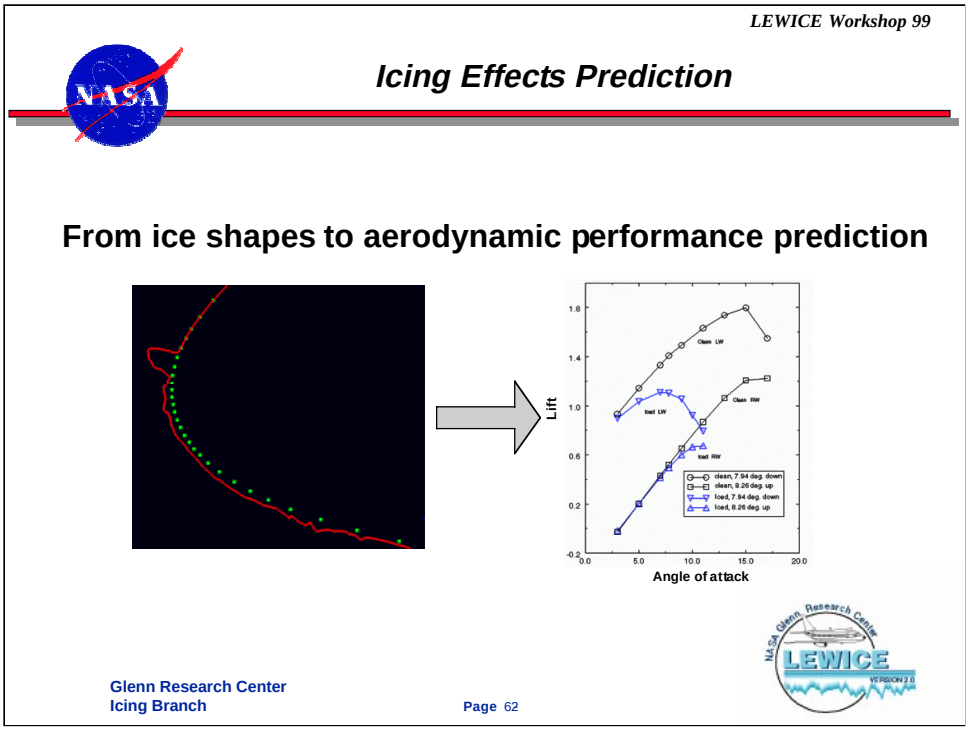
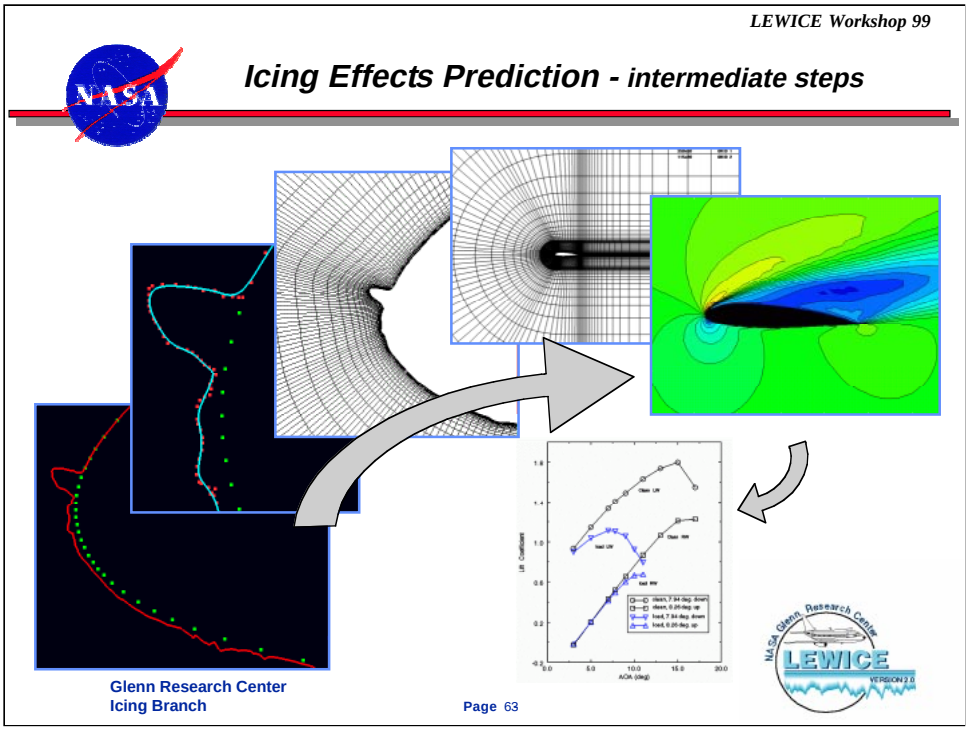
SmaggIce is a software tool developed at Glenn Research Center used to support the process of interactive prediction of icing effects on airfoils.



The problem

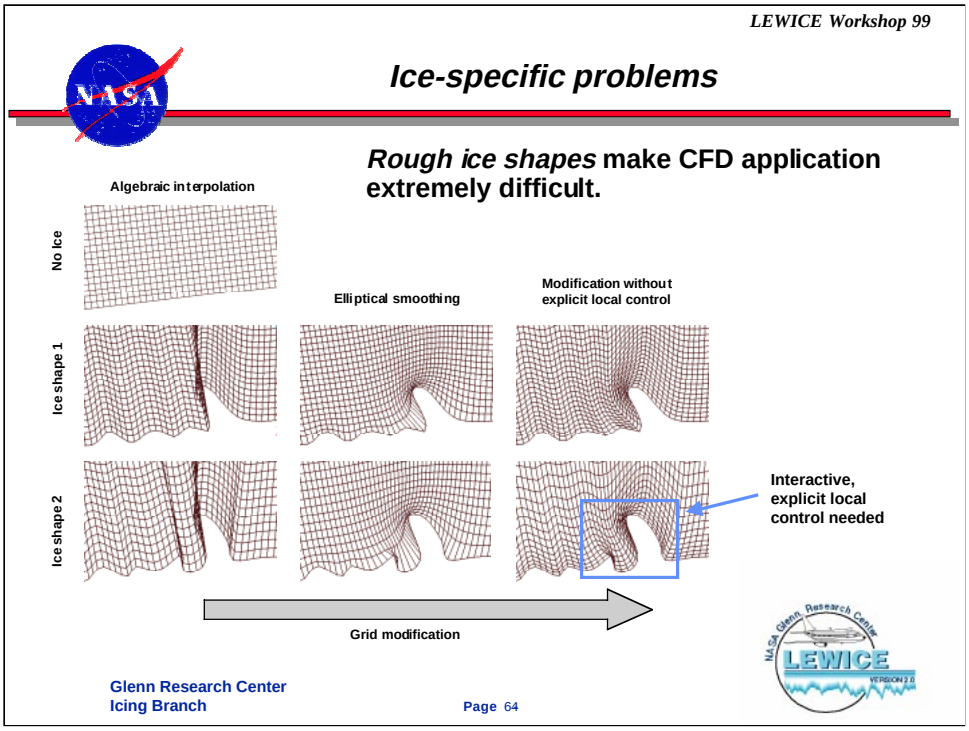
Given ice shapes (acquired either experimentally through icing tunnel or flight tests, or through numerical simulations), we need to determine the aerodynamic effects (e.g., lift and drag) of those ice shapes.

The image on the left shows a clean airfoil with ice accretion superimposed. Angle of attack vs. lift is plotted on the right. Compare results with and without ice accretion on a left and right airfoil.



Task details

- Acquire ice shape
- Characterize ice shape
- Prepare the surface for gridding
- Define the flow domain and generate grid
- Perform flow analysis using CFD tools
- Determine aerodynamic performance



Quality grids require smoothness, orthogonality, and resolution. With clean airfoils, we can generate grids to meet these requirements. But when ice is present, these requirements may conflict with one another. For example, smoothing may get rid of tangling, but not allow for proper resolution. Thus, application of grid-based CFD to iced airfoils is extremely difficult, if not impossible.

Global control can improve grid quality, but many ice shapes require explicit local control to generate quality grids that are needed to obtain accurate CFD solutions.

SmaggIce contribution

Phase 1

- **Surface preparation**
- **Measuring ice shape characteristics**

Phase 2

- **Gridding**
  - **domain decomposition**
  - **grid generation**
  - **grid quality check and control**

Phase 3

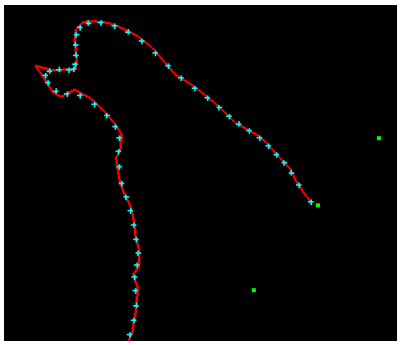
- **Tie to flow solver (WIND): input and feedback**
- **Demonstrate and test entire process**

Phase 3 will closely tie SmaggIce with the flow solver WIND*. Input to the flow solver is the grid, connectivity, and boundary conditions. If necessary, feedback from the solver will be used by SmaggIce to further modify the grid.

*Power, G.D., and M.L. Underwood, “WIND 2.0: Progress on an Applications-Oriented CFD Code”, AIAA Paper 99-3212, June 1999.

Phase 1 - Surface Preparation

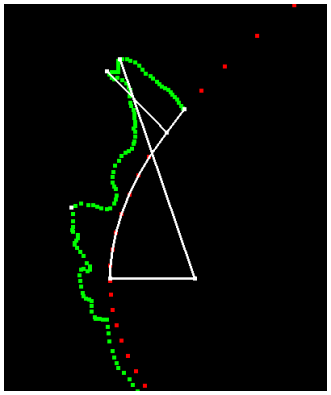
- Correcting input errors
- Surface smoothing
- Curve discretization
- Point redistribution
- Direct modification using control points



Phase 1 - Measuring Ice Shape Characteristics

Interactive Ice Shape Probing

- **Location**
- **Length**
- **Arc length**
- **Angle**
- **Relative X location**



Measurements are made interactively and may be normalized by the chord length of a reference airfoil.

SmaggIce Design Issues

- **Geometry**
  - **2D data**
  - **multiple ice shapes**
  - **multi-element airfoils**
- **UNIX platform independent**
  - **X Window system**
  - **OpenGL**
- **Dynamic memory**
- **Ease of use**
  - **error checking**
  - **guided use**

SmaggIce Status

- **Phase 1** - surface preparation, measurement
  - 1.0 beta available
  - 1.0 release - Feb. 2000
- **Phase 2** - gridding
  early 2001
- **Phase 3** - end-to-end process
  2002

1.0 Beta request forms available in lobby or send e-mail to: smaggice@grc.nasa.gov

Beta users will automatically receive 1.0 release.

Discussion

- **Are there any additional probing features that may be useful to your work?**
- **Can interactive ice-shape performance prediction be of value to your work? How?**